

CA-Software Assignment-1

Microservices & Event-Driven Architecture Scenario

A hypothetical e-commerce platform consists of independent services for User Management, Inventory, Orders, Payments, and Notifications. These services communicate via asynchronous messaging through an event bus.

SOLID Principles:

S (Single Responsibility): Each service focuses on one business capability.

O (Open/Closed): Services can be extended without modifying existing logic.

L (Liskov Substitution): Interfaces ensure consistent behavior across implementations.

(Interface Segregation): Services expose specific APIs for client needs.

(Dependency Inversion): Use of message brokers (e.g., Kafka) abstracts communication dependencies.

Bullet Points:

DRY: Shared logic (e.g., authentication) is isolated in common services.

KISS: Services remain simple, doing one job well.